Craig James Gleason III

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EDUCATION

University of Texas at Austin - B.S. in Electrical and Computer Engineering **Expected May 2028**

- **GPA:** 3.94
- **Relevant Courses:** Embedded Systems, Computer Architecture, Signals & Systems, Digital Logic Design

EXPERIENCE

Battery Management System (BMS) Lead **April 2026 - Present**
Longhorn Racing Solar **Austin, TX**

- Led **8** engineers in developing a **6-PCB** battery management system for a solar vehicle's **120 V, 40 A** pack, spanning cell monitoring, cell balancing, current monitoring, HV contactor control, and **CAN** telemetry
- Root-caused the failure that permanently destroyed last year's battery pack in storage, using bench current measurements to uncover a hidden **7 mA** parasitic drain from isolation circuitry, **20× higher** than expected
- Prevented a repeat failure by selecting **TI BQ79616** monitors with built-in isolated communication, eliminating the isolators behind the drain, and writing drivers that shut them down on comms loss, cutting storage draw to **6 µA** per board
- Cut active power draw **5×** versus the previous BMS, reaching **10 W active** and **25 µA quiescent** total system draw measured on bench, through contactor economizers, **isoUART** monitoring chips, and closed-loop fan control
- Reviewed all KiCad and firmware PRs across **8** repositories, catching schematic and driver defects before bring-up

Power Systems Member **September 2025 - March 2026**
Longhorn Racing Solar **Austin, TX**

- Developed **embedded C** drivers and an 8-task **FreeRTOS** architecture, using **priority isolation** to protect **safety-critical** monitoring and ISR-fed **CAN** queues to keep telemetry off the fault-response path, with **zero** on-track battery-related faults at competition (FSGP 2026)
- Designed a preemptive fault path that opens HV contactors in **under 7** ms via a highest-priority, lock-free task signaled by semaphore directly from detection logic, latching them open until a power cycle to prevent re-energization
- Wrote firmware test suites and a **GitHub Actions** CI pipeline that builds and runs them through Makefiles on every pull request, blocking merges that break the build or fail tests
- Administered the full BMS scrutineering test suite across **4 validation cycles** ahead of FSGP 2026, contributing to a **first-attempt pass** of electrical and BMS scrutineering
- Designed the BMS Leader Board in KiCad, which ran on the car at FSGP 2026: 24 V to 3.3 V buck stage with reverse-polarity and overvoltage protection, **STM32G4 MCU**, and I2C sensors (SHT45, EMC2025 fan controller)

UT Engineering College Coach **July 2026 - Present**
Harmony Public Schools **Austin, TX**

- Mentor **25** seniors from various Harmony high schools across Texas as they transition into UT's college of engineering, providing guidance, counseling, and advice

PROJECTS

FPGA Flight Computer **July 2026 - August 2026**

- Designed a flight computer PCB pairing an **iCE40 FPGA** with an STM32 MCU over **QSPI**, with the FPGA configured at boot from external NOR flash
- Wrote Verilog modules for **deterministic**, sub-millisecond arbitration of dual **I2C** IMUs on a fully open-source toolchain (Yosys, nextpnr, IceStorm)
- Developed STM32 firmware for servo control and two-tier logging: NOR flash for high-rate capture **(128 MB)** and an SPI SD card for bulk telemetry **(10 GB)**

R-Chess **December 2025**

- Wrote a complete terminal chess game from scratch in **Rust**, with full legal move generation, including castling, en passant, pawn promotion, as well as efficient check and checkmate detection
- Modeled board and move state around Rust's ownership rules, resolving borrow-checker and lifetime conflicts on shared mutable state, and debugged rule edge cases in castling rights, promotion operator precedence, and check detection

SKILLS

Embedded: Embedded C, Bare-metal & HAL, CAN, I2C, SPI, QSPI, UART, FreeRTOS, ARM, Verilog, GDB, SWD, Linux
Hardware: KiCad PCB Design & layout, Power Electronics, Soldering, Oscilloscopes, Logic Analyzer, STM32G4, ESP32-C6, iCE40
Software & Tools: C/C++, Rust, Python, CI/CD, Git, GitHub, STM32CubeMX, Makefiles